

LT Credit: Electric Vehicles

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core and Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses and Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To reduce pollution by promoting alternatives to conventionally fueled automobiles.

REQUIREMENTS

NC, CS, DATA CENTERS, HOSPITALITY, RETAIL, HEALTHCARE, SCHOOLS

Provide charging infrastructure for electric vehicles for on-site parking.

Option 1. Electric Vehicle Supply Equipment (1 point)

Install *electrical vehicle supply equipment (EVSE)* in 5% of all parking spaces used by the project or at least two spaces, whichever is greater. Clearly identify and reserve these spaces for the sole use by plug-in electric vehicles.

The EVSE must:

- ▶ Provide a Level 2 charging capacity (208 – 240 volts) or greater for each required space
- ▶ Comply with the relevant regional or local standard for electrical connectors, such as SAE Surface Vehicle Recommended Practice J1772, SAE Electric Vehicle Conductive Charge Coupler or IEC 62196 of the International Electrotechnical Commission for projects outside the U.S.
- ▶ Meet the *connected functionality* criteria for ENERGY STAR certified EVSE and be capable of responding to time-of-use market signals (e.g. price). Projects pursuing EA credit Grid Harmonization should incorporate EVSE into any demand response program or load flexibility and management strategies.

OR

Option 2. Electric Vehicle Ready Infrastructure (1 point)

Make 10% of parking spaces or at least 6 spaces, whichever is greater, *EV Ready*.

To be *EV Ready*, include a dedicated electrical circuit with sufficient capacity for each required space. Each circuit shall have conduit and wire sufficient to provide Level 2 charging or greater, and shall end at an electrical box or enclosure located near each required space.

SCHOOLS

Meet Option 1 or Option 2 above.

OR

Option 3: Electric buses or school-owned vehicles (1 point)

Develop and implement a plan for acquiring at least 1 electric bus and/or for every other bus serving the school to meet the following emissions standards within seven years of the building certificate of occupancy:

- ▶ nitrogen oxide (NOx) emissions of 0.50 grams or less per brake horsepower-hour; and
- ▶ particulate matter emissions of 0.01 grams or less per brake horsepower-hour.

Emission standards must be met for each bus and not by an average of the entire fleet serving the school.

Develop and implement a plan for 50% of all other (non-bus) vehicles owned or leased to serve the school to be electric vehicles.

WAREHOUSES & DISTRIBUTION CENTERS

Option 1. Electric Vehicle Charging (1 point)

Provide an on-site fleet with at least one *yard tractor* that is powered by electricity and provide on-site charging for the vehicles.

OR

Option 2. Reduced Truck Idling (1 point)

Provide an electrical connection for at least 50% of all dock door locations and/or truck parking locations to limit truck idling on the site.

Provide signage to communicate a reduced idling policy.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

This updated credit now refers to electric vehicles (EV) only, and the credit title has been changed to Electric Vehicles to reflect this. As the EV market grows around the world and outpaces other alternative fuels, updates are reflecting the huge potential of electric vehicles to contribute to a clean energy transition. Preferred parking requirements and other achievement barriers have also been removed and replaced with simplified language addressing newly added options that reward the installation of electric vehicle infrastructure.

Step-by-Step Guidance

- ▶ Disregard any references related to carpool preferred parking, alternative fueling, or permanently installed signage or pavement markings for preferred parking.
- ▶ Refer to all vehicles as electric vehicles.
- ▶ Projects are no longer required to meet the American Council for an Energy Efficient Economy (ACEEE) green score threshold of 45. However, project teams are encouraged to use the ACEE GreenerCars Ratings as a reference for information on vehicles ranked “Above Average.”
- ▶ Schools pursuing Option 3 are encouraged to collaborate with utility organizations and consider electric grid management options or electric bus storage opportunities. Working in conjunction with a utility company can help reduce stress on the electric grid during peak demand periods as well as help subsidize the cost of electric buses.

Further Explanation

Refer to the LEED v4 reference guide, with the following addition:

- ▶ Discounted parking rates are no longer applicable or viable as a substitute for preferred parking since it is no longer a requirement.
- ▶ Signage for preferred parking are no longer required. Signage for charging stations are still strongly encouraged, however.
- ▶ ACEEE scores no longer apply.

Required Documentation

Refer to the LEED v4 reference guide, with the following modification:

- ▶ Documentation related to preferred parking spaces, fueling stations, and discounted parking rates are no longer required.

Referenced Standards

- ▶ EPA ENERGY STAR Electric Vehicle Supply Equipment (EVSE) Key Product Criteria – Connected Functionality https://www.energystar.gov/products/evse_key_product_criteria

Definitions

- ▶ **EV Ready** A dedicated electrical circuit with appropriate capacity for an electric vehicle charging station.
- ▶ **electric vehicles (EV)** vehicles driven by electric motors which draw energy from either storage batteries or overhead cables.
- ▶ **electric vehicle supply equipment (EVSE)** the conductors, including the ungrounded, grounded, and equipment grounding conductors, the electric vehicle connectors, attachment plugs, and all other fittings, devices, power outlets or apparatuses installed specifically for the purpose of delivering energy from the premises wiring to the electric vehicle. (National Electric Codes and California Article 625).
- ▶ **connected functionality** refers to EVSE that are capable of supporting building and/or utility demand response programs (ENERGY STAR EVSE Key Product Criteria)

Connection to Ongoing Performance:

- ▶ LEED O+M LT prerequisite Transportation Performance: Providing and improving electric vehicle charging facilities and infrastructure implementing any corresponding electric vehicle incentivizing programs may help improve a project's transportation performance score.
- ▶ LEED O+M EA credit Grid Harmonization: Electric charging stations aligning with the existing demand response program or infrastructure comply with the related performance-based credit.